

Project

(CVA_Multicurve)

Financial Engineering

It is required to price an amortizing swap with the Counterparty risk computing the CVA term. Market volatilities are related to swaptions priced with a normal (Bachelier) model.

The Dataset

The dataset on the two reference dates 26th June 2022 (trade date) e 31st Jan 2023 (unwinding date), provided by Bloomberg and ICAP, includes:

- Deposit, Fras, STIR futures and Swaps market (vs Euribor 3m) quotes used to bootstrap curve in a multi curve framework;
- Swaptions normal Bachelier volatility matrices in bps for the strike of interest.

Euribor 3m normal (Bachelier) Volatility matrices are obtained from Bloomberg. On the 24th June 2022 is

The screenshot shows the Bloomberg VCBV (Interest Rate Volatility) interface. The main display is a volatility matrix for EUR IBOR, showing volatilities in bps for various maturities (1Yr to 30Yr) and strikes (1Mo to 30Yr). The interface includes a top menu bar with options like CANCEL, HELP, SEARCH, NEWS, QUOT, MSG, MENUL, PRINT, PG BA, and PG FV. Below the menu, there are tabs for Actions, Settings, and Market Data. The current view is set to 'Table' and 'Charts'. The matrix data is as follows:

Expiry	1Yr	2Yr	3Yr	4Yr	5Yr	7Yr	10Yr	12Yr	15Yr	20Yr	25Yr	30Yr
1Mo	199.84	212.56	198.77	184.19	171.97	159.20	140.45	137.54	134.39	129.24	124.00	119.97
3Mo	178.40	177.26	172.48	164.29	156.13	145.15	129.74	127.51	124.93	121.06	116.76	112.80
6Mo	154.56	156.09	151.99	145.75	139.10	130.17	118.87	117.41	115.76	112.80	109.41	106.18
9Mo	143.08	147.30	141.40	135.26	130.06	123.36	113.64	112.54	111.24	108.64	106.58	103.39
1Yr	133.81	137.43	132.88	126.90	122.45	116.57	108.07	107.36	106.52	104.35	101.92	99.56
2Yr	121.09	121.52	116.67	113.38	109.72	104.93	98.82	97.87	96.48	95.11	92.55	90.15
3Yr	112.31	111.59	107.65	104.33	101.41	97.49	92.22	91.17	89.55	88.15	85.51	82.83
4Yr	105.23	104.38	101.77	98.64	95.86	92.37	87.97	86.68	84.66	83.08	80.28	77.37
5Yr	99.18	97.92	95.61	93.21	91.29	88.44	84.75	83.34	81.12	79.62	76.93	74.13
6Yr	94.13	93.09	90.57	88.38	86.65	84.44	81.35	79.93	77.70	76.24	73.66	71.03
7Yr	89.66	88.76	86.31	84.39	82.66	80.75	78.16	76.79	74.64	73.09	70.54	68.11
8Yr	86.31	85.79	83.72	81.81	80.34	78.38	75.72	74.38	72.30	70.45	68.08	65.88
9Yr	83.42	83.31	81.36	79.84	78.39	76.47	73.31	72.01	70.00	68.03	65.89	63.86
10Yr	81.53	81.33	79.80	78.15	76.55	74.61	70.99	69.74	67.82	65.87	63.81	61.97
12Yr	78.34	78.18	76.63	75.80	74.40	72.11	68.41	67.04	64.95	62.76	60.93	59.14
15Yr	76.42	76.01	74.37	72.86	71.15	68.79	64.86	63.27	60.84	58.56	56.80	55.01
20Yr	72.01	71.79	70.27	68.67	67.12	64.69	60.68	59.03	56.54	53.83	51.88	49.85
25Yr	69.86	69.64	67.95	66.26	64.49	61.51	57.25	55.49	52.84	49.74	47.37	45.16
30Yr	67.41	67.47	65.86	64.00	62.10	58.91	54.33	52.35	49.37	46.34	43.88	41.61

At the bottom of the interface, there are sections for 'Quick Pricer For Swaption', 'NSV Monitor swaption volatility in real time', and 'OTC Track performance in an emerging market'.

and on the 31st Jan 2023 is

VCUB Interest Rate Volatility

CANC HELP SEARCH NEWS QUOT QUOT MSG MENU PRINT PG BA PG FV

< > DIAMONDBACK ENER Equity | VCUB | Related Functions Menu

<Back> to Return

92 Actions 93 Settings Interest Rate Volatility Cube

EUR IBOR EUR BVOL Cube (Default) Mid Date 01/31/23

99 Analyze Cube 90 Market Data

Swap Curve (45) Euro Index Tenor 6M Show Vol Normal

View Strike 2.21 Discounting OIS Show Strikes

Table Charts

Expiry	1Yr	2Yr	3Yr	4Yr	5Yr	7Yr	10Yr	12Yr	15Yr	20Yr	25Yr	30Yr
1Mo	94.86	112.59	111.78	112.44	113.67	114.22	114.82	114.86	114.87	111.97	106.54	102.40
3Mo	92.16	105.61	104.57	105.74	107.19	106.44	105.76	105.56	105.22	103.83	101.15	99.38
6Mo	91.98	99.38	99.17	100.26	101.75	101.24	101.10	101.11	101.07	99.57	98.08	96.94
9Mo	87.52	93.94	95.79	97.29	98.41	99.06	98.81	98.48	97.93	95.86	94.61	93.18
1Yr	86.60	92.45	94.25	95.42	96.56	97.61	97.23	96.74	95.95	93.56	92.06	91.01
2Yr	91.24	94.18	94.66	94.91	94.97	95.01	95.00	93.82	91.98	90.30	88.41	87.10
3Yr	93.22	94.34	93.72	93.27	93.08	92.56	91.69	90.41	88.42	86.29	84.59	83.04
4Yr	92.82	93.14	92.11	91.51	91.05	90.02	88.24	86.74	84.42	82.20	80.63	78.95
5Yr	91.17	91.04	90.21	89.27	88.64	87.13	84.93	83.31	80.80	78.43	76.70	75.07
6Yr	89.09	88.98	87.79	86.86	86.43	84.76	82.28	80.57	77.95	75.50	73.82	72.05
7Yr	87.25	86.94	85.94	84.62	84.06	82.25	79.97	78.21	75.54	72.91	71.16	69.48
8Yr	85.49	85.46	83.93	82.97	82.20	80.43	78.14	76.27	73.46	70.49	68.77	67.14
9Yr	84.21	84.24	82.49	81.47	80.41	78.67	76.25	74.24	71.24	68.16	66.48	64.85
10Yr	83.10	83.16	81.59	80.15	78.77	77.00	74.29	72.17	69.06	65.98	64.24	62.69
12Yr	81.35	81.17	79.48	78.15	76.66	74.52	71.22	69.03	65.79	62.96	61.20	59.63
15Yr	79.97	79.70	77.64	75.64	73.33	70.84	67.21	64.86	61.36	58.98	57.07	55.33
20Yr	77.60	77.64	75.17	72.60	70.00	67.13	62.95	60.51	56.83	54.29	52.60	50.72
25Yr	74.16	74.18	71.60	68.88	66.17	62.89	58.71	56.40	52.95	49.92	48.23	46.55
30Yr	71.91	71.94	69.18	66.40	63.23	59.78	55.03	52.62	49.00	46.15	44.59	43.06

Quick Pricer For Swaption Swap Manager (SWPM)

Suggested Functions IRSB Spot opportunities in the IRS market FXFC Evaluate FX market forecasts

1. Multicurve calibration

For the multi-curve bootstrap consider the paper [1]. Neglect the convexity adjustment term in STIR futures.

2. Risk Free Amortizing Swap Pricing

Determine the $NPV_{RiskFree}$ of the swap in the Annex with the amortizing plan in SwapAmortizingPlan.xlsx and assuming absence of counterparty risk (e.g. due to the presence of a CSA). Formulas are analogous to the single curve model.

Hint 1:

Pay attention that in a multicurve model, the telescopic sum relation for the floating leg of a swap does not hold anymore.

3. Amortizing Swap Pricing with CVA: simplified approach.

Compute the Net present value for the swap in the Annex at Trade Date.

Consider

$$NPV = NPV_{RiskFree} - CVA$$

where the CVA (cfr. formula 5.5 in [3] o 4.13, pag.98 in [4]) is

$$CVA = LGD \sum_{i=1}^{b-1} (Q\{\tau > t_{i-1}\} - Q\{\tau > t_i\}) PS_{i,b}(t_0; K, \hat{S}_{i,b}(t_0), \hat{\sigma}_{i,b})$$

where LGD=40% and consider 2 scenarios for the -constant in time- CDS Spread (300bp and 500bp).

For the payer swaption $PS_{i,b}(t_0; K, \hat{S}_{i,b}(t_0), \hat{\sigma}_{i,b})$ consider the normal (Bachelier) formula in the multi-curve setting with OIS discounting with the corresponding amortizing fwd BPV $\widehat{BPV}_{\alpha\omega}(t_0)$ and the amortizing forward rate $\hat{S}_{\alpha\omega}(t_0)$. In the amortized case the fwd-BPV is:

$$\widehat{BPV}_{\alpha\omega}(t_0) = \sum_{i=\alpha+1}^{\omega} \frac{N_i}{N_{\alpha}} \delta(t_{i-1}, t_i) B(t_0; t_{\alpha}, t_i)$$

where:

N_i :	Amortizing notional in the time interval (t_{i-1}, t_i)
$\delta(t_{i-1}, t_i)$:	Year frac in the time interval (t_{i-1}, t_i) ; Act/360
$B(t_0; t_{\alpha}, t_i)$:	Forward discount on the OIS curve

The volatility for the swaption in the amortized case can be obtained through a fwd-BPV interpolation from the plain vanilla ones. Consider the options with same expiry t_{α} ; the option with maturity t_j of the underlying swap has $BPV_{\alpha j}(t_0)$.

Comment the results.

Hint 2: First, you should linearly interpolate the volatilities with expiry t_{α} and for all tenors of interest τ_i obtaining $\hat{\sigma}_{\alpha,i}$. Then, among the options with same expiry t_{α} , you should linearly interpolate in terms of the BPV the volatility $\hat{\sigma}_{\alpha,b}$ for the amortizing swap of interest.

4. Unwinding

The corporate unwinds the swap on the 31st Jan 2023, i.e. it closes the swap with the banking counterparty paying the NPV at that date.

Assuming that the credit risk of the corporate remains the same on the 31st Jan 2023, what is the amount that should be required by the Bank?

Comment the results.

5. Calibration Multicurve Swaption model for some given values of γ

Calibrate the model described in [2] for the two reference dates.

- Use only the “diagonal” swaption values (1y15y, 3y12y, 5y10y, 8y7y, 10y5y, 12y3y, 15y,1y)
- Calibrate the parameters a & b considering three possible (fixed) values for γ : 0, $\frac{1}{2}$ and 1.

6. [Facultative] Amortizing Swap Pricing with CVA with numerical technique.

Price the NPV of the swap at start date for $\gamma = 0$ via a numerical technique.

Generalize to this case the Hull-White tree discussed at lesson. Discuss the results.

Hint 3:

The tree for the OU construction can be used even in this multi-curve setting, paying attention to the relation between the OU process and the quantities of interest in the multi-curve model.

Realize a library in Matlab. Optional Python.

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Bibliography

- [1] Baviera, R., & Cassaro, A. (2015). A note on dual-curve construction: Mr. Crab's Bootstrap. *Applied Mathematical Finance*, 22(2), 105-132.
- [2] Baviera, R. (2019). Back-of-the-envelope Swaptions in a Very Parsimonious Multi-Curve Interest Rate Model. *International Journal of Theoretical and Applied Finance*, 22(05), 1950027.
- [3] Brigo, D., & Masetti, M. (2005). Risk neutral pricing of counterparty risk, 1-39, In *Counterparty Credit Risk Modeling: Risk Management, Pricing and Regulation*. Risk books, London.
- [4] Brigo, D., Morini, M., & Pallavicini, A. (2013). Counterparty credit risk, collateral and funding: with pricing cases for all asset classes (Vol. 478). John Wiley & Sons. Ch. 4.

Swap Annex:

Indicative Terms and Conditions as of 26th June 2022

Swap Termsheet

Principal Amount (N):	20 MIO EUR
Amortizing plan:	See annex
Trade Date:	24 th June 2022
Settle Date:	28 th June 2022
Maturity Date:	28 th June 2037 (15y), subject to the Modified Following BD convention.
Reset Dates:	2 BD prior to each semiannually calculation start date
Bank payment dates:	Quarterly, subject to Following BD convention
Daycount:	Act/360
Bank receives:	Euribor 3M

Corporate payment dates:	Quarterly, subject to Following BD convention
Corporate receives:	2.21%
Daycount:	Act/360